

seen that start-ups and SMEs tend to focus more on technology.

Let me give an example from some Japanese product companies. Even high-end electronics products from Japanese vendors, if you look at their items like power supplies and other low-tech components, use older technologies like through-hole parts, paper phenolic PCBs, etc.

On the contrary, recently I reviewed a start-up's PCB designed for a controller. The enclosure was big, but they had designed an extremely small PCB with small-footprint SMT (surface mount technology) parts. They were struggling with the design even after three revisions.

I stepped in because their investor asked me to review. I told them change to through-hole parts and use a simple two-layer board for the circuitry. They resisted citing SMT as the current technology. But their investor gave them no option, so they developed a 'low-tech' board reluctantly. As expected, the design worked. Their software team could get the very first version of the product working.

It is not 'high-tech' that matters, it is 'appropriate tech' that matters. Had they done their planning properly, it would have given them the cue to use a simpler technology and they need not have wasted time.

Most of the times when you develop a new product you have to get your product tested in the field before going for large-scale manufacturing. Most start-ups and SMEs assume the prototype can be quickly converted to a regular product, with little effort, which is a fallacy. Converting a prototype or an early-stage version to a manufacturable product takes lots of effort, time, and money.

Designers need to understand that EMS vendors do not have engineering teams to completely engineer the product for volume manufacturing. All this calls for good planning, without which a product will not be successful.

Some examples

A start-up had a novel design and developed the prototype with open source hardware components. As part of their pitching, they presented it to their prospective customers. One of

them wanted a pilot trial for the product which used IoT hardware. The customer was willing to pay for the pilot as he saw value in the solution. But when the start-up tried building 50+ prototypes they could not as they had used parts which were not easily available at a regular price. My advice has been that for proving an idea for a few units' testing, open source components are good but not for serious designs.

Then there was an SME who developed a probe for measuring battery voltage in a battery production line. Basically, in the manufacturing line a bunch of cells are connected together and charged. Voltages of the cells are normally measured manually, which was taking time. The battery manufacturer wanted an automated solution. The SME designed a unit and made fifty prototypes. But when it was deployed, no probe was giving the 1mV accuracy the battery manufacturer wanted. So, after a month the trial was stopped.

The battery vendor requested me to analyse what had gone wrong. (Their worry was whether there was any hidden issue in their manual measurement process?) When I reviewed the design, I found the design was basically done for hand-crafting and was not suitable for large-volume production. Primary issues were: (a) there were no provisions in the design for periodic calibration of the probes to meet the desired accuracy, and (b) it was not engineered for manufacturing. Had the team done detailed planning this project would have been successful.

What needs to be planned?

Planning for hardware products is done to ensure all the activities including design, manufacturing, certification, and field installation are covered and suitable actions are planned at the start of the project. Some of the critical items are:

1. Components for the final manufacturing and the sources to be identified. This ensures the lead times are within reasonable limits.

2. Selection of manufacturing partner, components vendors for parts like sheet metal, plastics, tooling, moulding, cable harness, and third-party

hardware supporter if you want to deploy in larger geography.

3. Planning for product certification upfront and the agency who will do it. Planning at the end can create huge time and technical pressure.

4. Ensure the target price is under the radar and product design adheres to it.

Tips for planning

Let me give a few important tips for the start-ups and SMEs to keep the planning simple.

1. Complete the entire solution (hardware, software, cloud, etc) in one go; do not do the designing in bits and pieces.

2. Identify partners for EMS, tooling, packaging, etc early on, preferably during the design phase, and firm up the relationships.

3. Price and volume are interconnected. So be aware that initial low volumes will cost high. This essentially means start-ups and SMEs have to call on what is the best price client is willing to pay. Typically, if your solution is going to save cost for the end customer, look at a subscription based business model which allows you to recoup the cost of hardware and margins through subscriptions.

4. Understand the standards the product needs to comply with. With India integrating into the global market, local compliance standards are getting harmonised with the global standards. Designing a standards complying product costs money and effort. Plan that well early.

5. If you launch a new product, remember the software might have to be upgraded as the bugs are gradually getting fixed. True over-the-air (OTA) update will call for a controller with large memory and reliable link. This will make the product expensive initially. Plan for this too. In case cost does not allow an OTA-compliant hardware, devise a robust mechanism to provide upgrade for the software. This calls for good planning in the beginning itself. **EFY**



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